

Assignment 3: Naive Bayes Spam Classifier

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Setup

The dataset has 30 text messages labeled spam or ham. Following the instructions, the first 20 messages are the training set and the last 10 are the testing set. Each message is lowercased, punctuation is removed, and it is split into words. The training vocabulary has $|V| = 158$ unique words. Spam training messages have 166 total words and ham training messages have 83.

Task 2: Prior Probabilities

Out of the 20 training messages, 9 are spam and 11 are ham.

	Count	Formula	Prior
P(spam)	9	9 / 20	0.45
P(ham)	11	11 / 20	0.55

Conditional Probabilities (Laplace smoothing)

$P(\text{word}|\text{spam}) = (\text{count in spam} + 1) / (166 + 158)$ and $P(\text{word}|\text{ham}) = (\text{count in ham} + 1) / (83 + 158)$.

A few examples from the training set:

Word	P(word spam)	P(word ham)
call	0.0216	0.0083
free	0.0154	0.0041
prize	0.0093	0.0041
you	0.0185	0.0332
i	0.0031	0.0249
am	0.0031	0.0166

Test Set Results

#	Sentence	P(spam s)	P(ham s)	Predicted	Actual
1	Tell where you reached	0.0859	0.9141	ham	ham
2	Your gonna have to pick up a burger for yourself on your way home	0.3281	0.6719	ham	ham
3	As a valued customer I am pleased to advise you that for your recent review you are awarded a Bonus Prize	0.1095	0.8905	ham	spam
4	Urgent you are awarded a complimentary trip to EuroDisinc To claim text immediately	0.9406	0.0594	spam	spam
5	Finished class where are you	0.1226	0.8774	ham	ham
6	where are you how did you perform	0.0036	0.9964	ham	ham
7	you can call me now	0.2684	0.7316	ham	ham
8	I am waiting Call me once you are free	0.0154	0.9846	ham	ham

9	I am on the way to homei	0.0397	0.9603	ham	ham
10	Please call our customer service representative between 10am-9pm as you have WON a guaranteed cash prize	0.9532	0.0468	spam	spam

Posteriors are normalized so $P(\text{spam}|s) + P(\text{ham}|s) = 1$. The highlighted row is the one wrong prediction.

Task 6: Test Set Accuracy

Accuracy = correctly predicted / total = $9 / 10 = \mathbf{0.90 (90\%)}$

The one miss was test sentence 3, a spam message the model labeled ham. It contains "I" and "am", which only appear in ham training messages, and seven of its words (like "awarded" and "bonus") never appear in training at all. Because ham has fewer total training words, an unseen word gets a slightly higher smoothed probability under ham ($1/241$ vs $1/324$), so all those unseen words added up and pushed the result toward ham.

AI Use Disclosure

I used Claude (an AI assistant) to help me with this assignment. I came in understanding the big picture of Naive Bayes from lecture, and I used Claude to help write the Python code and walk me through each step so I understood why it works, not just that it runs.

I went through it step by step. First I split the 30 messages into 20 for training and 10 for testing. Then I found the priors by counting how many training messages were spam (9) and ham (11). After that I worked through how $P(\text{word}|\text{spam})$ and $P(\text{word}|\text{ham})$ are counted, and why Laplace smoothing adds 1 to every count so a word we never saw doesn't zero out the whole sentence. Then I followed how the naive assumption lets us multiply the word probabilities together, and how multiplying that by the prior gives the posterior for each class.

The part that clicked most for me was test sentence 3. It's spam, but the model called it ham. When I looked into why, it was because "I" and "am" show up a lot in the ham training messages, and seven of its words (like "awarded" and "bonus") never showed up in training at all. Since ham has fewer total words, those unseen words get a slightly higher smoothed probability under ham, which pushed the answer the wrong way. Seeing that showed me how the formula actually affects the final decision.

Claude also helped with the wording of the comments and the report. I reviewed all of the code and the results, and I can explain each step of how the model decides what is spam and what isn't.